

Open Ended Problem-7

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Code:-

```
import cv2

import matplotlib.pyplot as plt

import numpy as np


image1 = cv2.imread("image1.png", 0)
image2 = cv2.imread("image2.png", 0)
image3 = cv2.imread("image3.png", 0)
image4 = cv2.imread("image4.png", 0)
image5 = cv2.imread("image5.png", 0)


# Increase brightness
bright_image = cv2.add(image1, 50)


# Find minimum and maximum pixel values
min_value = np.min(image2)
max_value = np.max(image2)


# Apply contrast stretching
contrast_image = ((image2 - min_value) /
```

```

(max_value - min_value) * 255)

contrast_image = contrast_image.astype(np.uint8)

# Convert image to float
image3_float = image3.astype(float)

# Log transformation
c = 255 / np.log(1 + np.max(image3_float))

log_image = c * np.log(1 + image3_float)

log_image = log_image.astype(np.uint8)

equalized_image = cv2.equalizeHist(image4)

# Use image4 as the reference image
reference_image = image4

def histogram_matching(source, reference):

    source_hist = np.histogram(
        source.flatten(),
        bins=256,
        range=(0, 256)

```

```
)[0]
```

```
reference_hist = np.histogram(  
    reference.flatten(),  
    bins=256,  
    range=(0, 256)
```

```
)[0]
```

```
# Calculate CDF
```

```
source_cdf = np.cumsum(source_hist)  
reference_cdf = np.cumsum(reference_hist)
```

```
# Normalize CDF
```

```
source_cdf = source_cdf / source_cdf[-1]  
reference_cdf = reference_cdf / reference_cdf[-1]
```

```
# Create mapping
```

```
mapping = np.zeros(256, dtype=np.uint8)
```

```
for i in range(256):
```

```
    difference = np.abs(reference_cdf - source_cdf[i])  
    mapping[i] = np.argmin(difference)
```

```
# Apply mapping
```

```
matched_image = mapping[source]
```

```
return matched_image
```

```
matched_image = histogram_matching(image5, reference_image)
```

```
plt.figure(figsize=(12, 10))
```

```
plt.subplot(5, 2, 1)
```

```
plt.imshow(image1, cmap="gray")
```

```
plt.title("Image 1 - Original")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 2)
```

```
plt.imshow(bright_image, cmap="gray")
```

```
plt.title("Brightness Transformation")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 3)
```

```
plt.imshow(image2, cmap="gray")
```

```
plt.title("Image 2 - Original")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 4)
```

```
plt.imshow(contrast_image, cmap="gray")
```

```
plt.title("Contrast Stretching")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 5)
```

```
plt.imshow(image3, cmap="gray")
```

```
plt.title("Image 3 - Original")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 6)
```

```
plt.imshow(log_image, cmap="gray")
```

```
plt.title("Log Transformation")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 7)
```

```
plt.imshow(image4, cmap="gray")
```

```
plt.title("Image 4 - Original")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 8)
```

```
plt.imshow(equalized_image, cmap="gray")
```

```
plt.title("Histogram Equalization")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 9)
```

```
plt.imshow(image5, cmap="gray")
```

```
plt.title("Image 5 - Original")
```

```
plt.axis("off")
```

```
plt.subplot(5, 2, 10)
```

```
plt.imshow(matched_image, cmap="gray")
```

```
plt.title("Histogram Matching")
```

```
plt.axis("off")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:

Image 1 - Original



Image 2 - Original

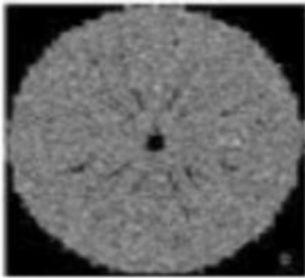


Image 3 - Original

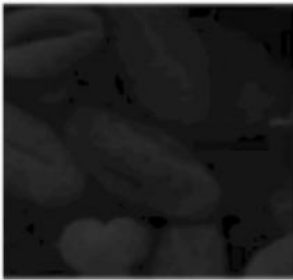
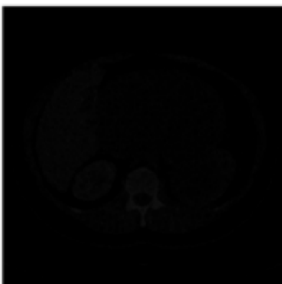


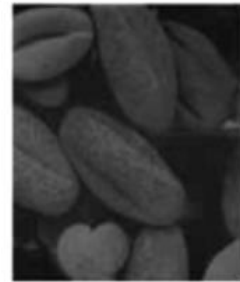
Image 4 - Original



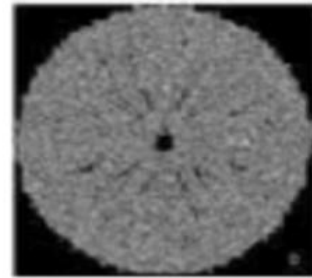
Image 5 - Original



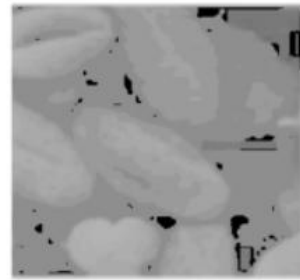
Brightness Transformation



Contrast Stretching



Log Transformation



Histogram Equalization



Histogram Matching



Objective

1. To enhance the visual quality of five degraded grayscale images using suitable gray-level transformation techniques.
2. To apply brightness transformation, contrast stretching, logarithmic transformation, histogram equalization, and histogram matching.
3. To improve image brightness, contrast, and visibility of hidden details.
4. To compare the effectiveness of different gray-level enhancement techniques.
5. To study the advantages and limitations of each enhancement technique.

Conclusion

The experiment successfully enhanced five degraded grayscale images using different gray-level transformation techniques. Brightness transformation improved dark images, contrast stretching enhanced the intensity range, and logarithmic transformation revealed details in darker regions. Histogram equalization improved overall contrast, while histogram matching adjusted the intensity distribution according to a reference image. Each technique was effective for a specific type of degradation, but may also have limitations such as overexposure, noise amplification, or unnatural contrast. Thus, the appropriate gray-level transformation should be selected based on the type of degradation in the image.